

Swapnil Upadhyay

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📄 PROFESSIONAL SUMMARY

Computer Science graduate proficient in predictive modeling, statistical analysis, and real-time dashboarding (Python, SQL, Scikit-Learn, PostgreSQL, Streamlit). Demonstrated success building end-to-end data applications (*MarkIt*) that process unstructured vision and audio data and deliver automated, high-precision analytics.

🏢 PROFESSIONAL EXPERIENCE

Software Development Intern, *Telite Systems Pvt. Ltd.* 02/2026 – 07/2026 | Tehri Garhwal, Uttarakhand

- **Data Processing & Backend Pipelines:** Engineered backend REST APIs and asynchronous data ingestion pipelines for an enterprise **Learning Management System (LMS)** using Python 3.12 (FastAPI), SQLAlchemy, and PostgreSQL.
- **Automated Analytics & Reporting:** Leveraged Pandas, openpyxl, Celery, and Redis to automate background batch data aggregation, feature reporting, and scheduled metric generation for user analytics.
- **Data Architecture & Infrastructure:** Designed structured relational schemas utilizing PostgreSQL Row-Level Security (RLS) for multi-tenant data isolation and containerized services using **Docker Compose** for reproducible data environments.

🎓 EDUCATION

Bachelor of Technology in Computer Science and Engineering 2022 – 2026 | Tehri Garhwal, Uttarakhand

THDC Institute of Hydropower Engineering and Technology

CGPA: 7.9

🧠 TECHNICAL SKILLS

- **Programming:** Python, SQL, HTML5, Custom CSS, Streamlit
- **Data Analysis:** Pandas, NumPy, Statistics, Data Cleaning, Exploratory Data Analysis (EDA)
- **Databases:** Supabase, PostgreSQL, Relational DB Design, bcrypt Hashing, Authentication, Access Control (RBAC)
- **Machine Learning:** Computer Vision, Face Recognition, Deep Learning, Biometrics, Support Vector Machines (SVM), Scikit-Learn, dlib, Feature Embedding, Euclidean Distance
- **Tools:** Jupyter Notebook, Git, GitHub, Docker, Pandas, NumPy, OpenCV, segno (QR Code), Git, CMake

📁 PROJECTS

India Public Health Analytics 📄, *Python (Pandas, SciPy, Statsmodels, Seaborn)*

- **Data Cleaning & EDA:** Built a Python pipeline to clean survey data across 637 Indian districts (96 indicators) with automated domain validation to analyse regional healthcare disparities.
- **OLS Regression Finding:** Trained an OLS regression model ($R^2=0.5831, p<0.0001, VIF<10$), proving **4+ Antenatal Care visits** ($\beta=+0.34$) and **female literacy** ($\beta=+0.28$) drive **58.3%** of institutional birth rate variance.
- **Hypothesis Testing Finding:** Executed T-Tests and ANOVA in SciPy ($p<0.001, p<0.001$), revealing high-literacy districts achieve **+25.5%** higher hospital birth rates (**84.2%** vs **58.7%**) and Central/Eastern districts suffer severe child anaemia (>62%) vs Southern/Western (<48%).

Mark It - AI Powered Attendance System 📄,

Python, Streamlit, Supabase (PostgreSQL), Computer Vision (dlib), Deep Audio Processing (Resemblyzer), Scikit-Learn, Bcrypt

- **Engineered an end-to-end computer vision pipeline** using dlib's 68-point facial landmark pose predictor and the ResNet face recognition model to extract 128-D facial feature embeddings from single-person and multi-person classroom photos.
- **Developed a voice biometric verification module** utilizing Resemblyzer's deep neural network encoder to extract 256-D speaker embeddings, then matching utterance samples against enrolled voice templates via dot-product cosine similarity.
- **Optimized inference performance & latency** by implementing Streamlit's `@st.cache_resource` memory caching for model loading and SVM fitting, preventing redundant recalculations during multi-photo classroom scans.

🏆 NOTABLE ACHIEVEMENTS

GATE-2026 Qualified, *GATE 2026(CS)*

- Qualified the GATE-2026 Exam with an overall score of 367.

Inter-Collegiate Football Tournament, *Runner-Up*

- Represented the college football team in the University sports meet and secured the runner-up trophy.